



PLANT LEAF DISEASE DETECTION

P R E S E N T A T I O N
B Y L E A F I T T O U S





OUR TEAM



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Introduction

Business Understanding

- According to the World Bank (2023), agriculture employs about 65% of Africa's workforce.
- The FAO (2019) reports that plant diseases cause 20–40% of global crop losses, worth over \$220 billion yearly.
- In Africa, outbreaks like tomato blight and maize rust can reduce yields by up to 50% (CIMMYT, 2021).
- Most farmers still rely on manual inspection, which is slow and error-prone.
- This project uses machine learning (CNN) to detect leaf diseases early and accurately.



Problem Statement

Farmers often struggle to identify plant diseases early enough to prevent damage.

Manual methods are slow and prone to errors

Existing detection methods are costly or unavailable in remote areas.

There is a need for a low-cost, accessible, and accurate solution to identify plant diseases using images.

Our Goal:

To develop a simple, accurate, and efficient image-based system for early detection of plant leaf diseases.



Objectives

Main Objective

To build a reliable model that classifies plant leaf images as healthy or diseased.

Specific Objectives

- Identify and classify diseases in corn, tomato, and potato leaves.
- Improve accuracy and reduce misdiagnosis.
- Support farmers and agricultural officers in timely interventions.



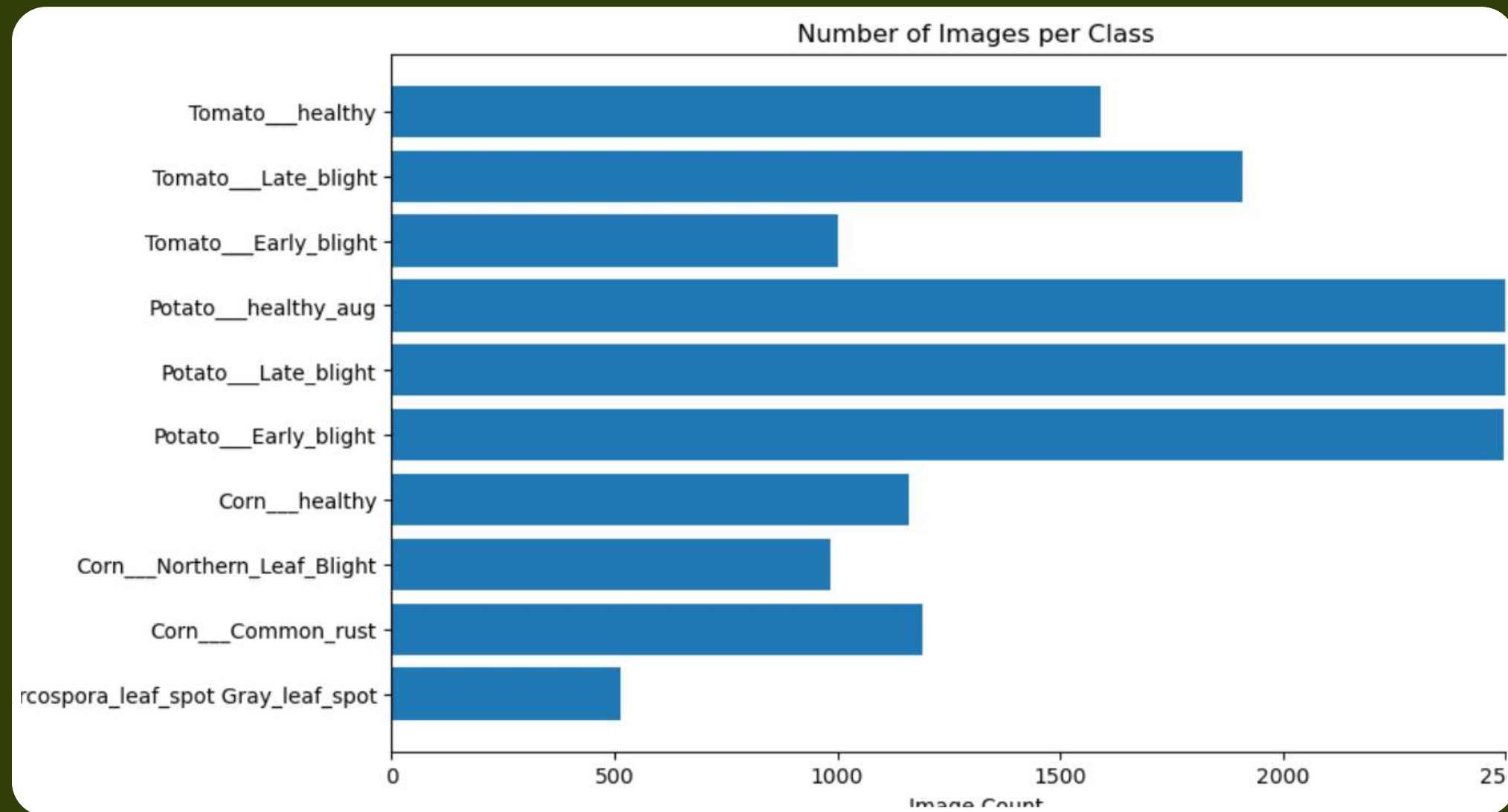
Methodology



- **Data Loading:** Imported and organized all plant images by class.
- **Data Cleaning:** Removed duplicates and corrupted images.
- **Exploratory Data Analysis (EDA):** Checked class distribution and sample visuals.
- **Data Preparation:** Split dataset into training (70%), validation (15%), and testing (15%).
- **Modelling:** Trained CNN and transfer learning models.
- **Evaluation:** Compared model accuracies and tested final deployment.

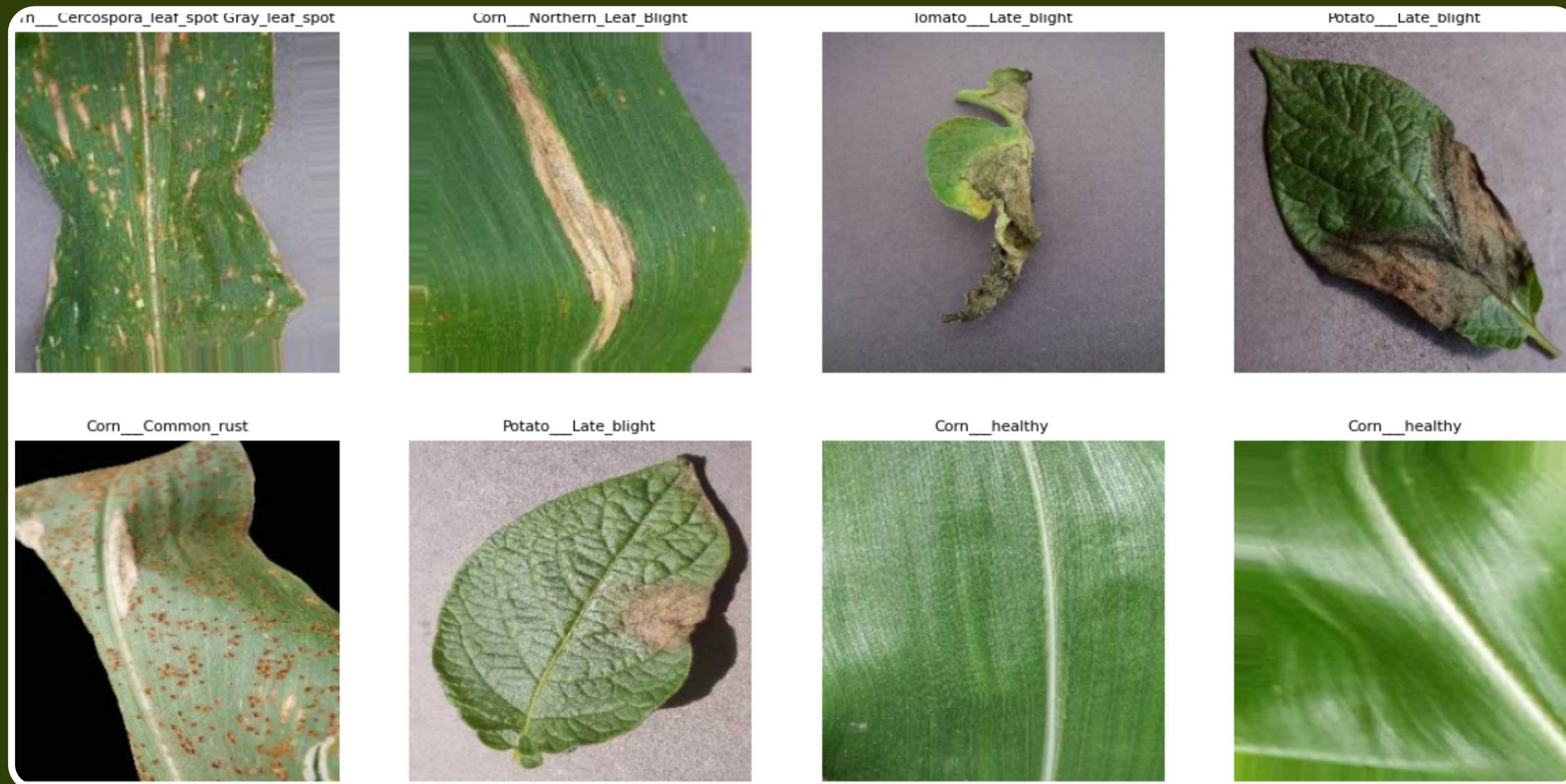
Data Understanding

- Dataset includes images of corn, tomato, and potato leaves (healthy and diseased).
- Sourced from PlantVillage dataset and similar open agricultural repositories.
- Each image labeled by crop and disease type.
- Data split into training, validation, and test sets.



Data Understanding

- We had a total of 10 classes: Corn Healthy, Corn Northern Leaf Blight, Corn Common Rust, Corn Gray Leaf Spot, Tomato Healthy, Tomato Late Blight, Tomato Early Blight, Potato Healthy, Potato Late Blight, Potato Early Blight



Data Analysis

Exploratory Insights:

- Visualized sample images per class to confirm balance and diversity.
- Checked image dimensions and quality consistency.
- Identified imbalance, some diseases had fewer samples.

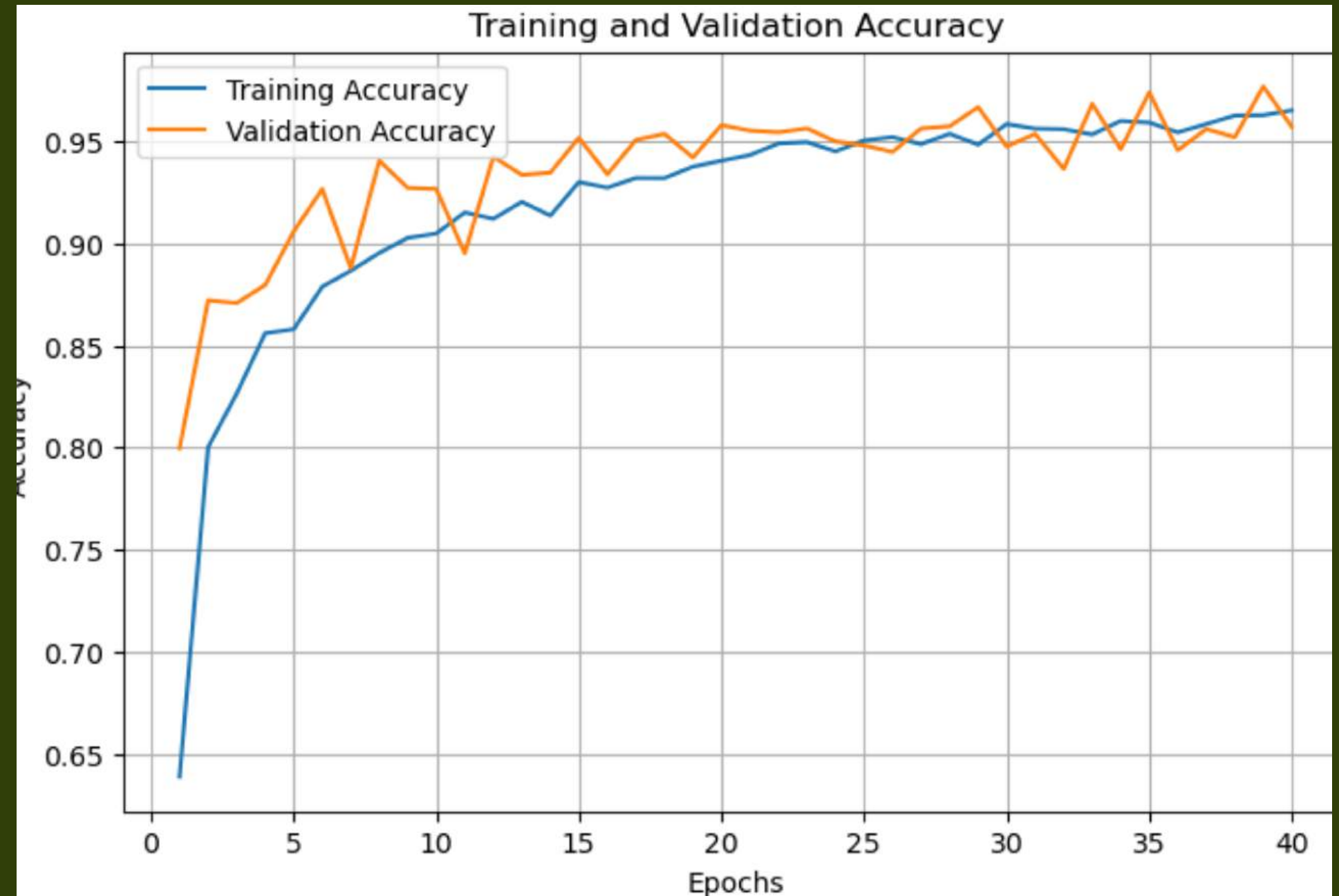
Solution

- Applied ImageDataGenerator for data augmentation.
- Techniques: rotation, zooming, flipping, and rescaling



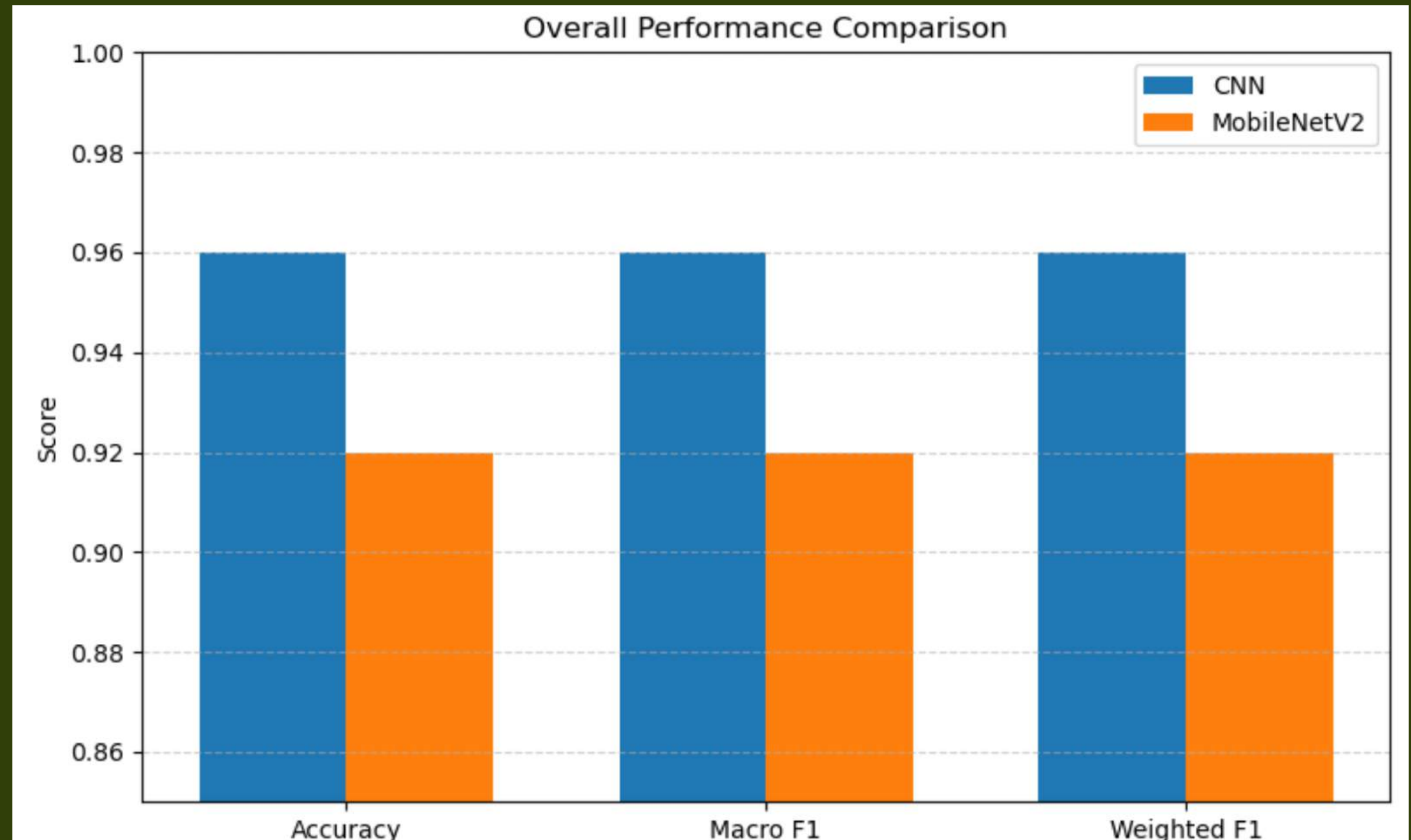
Modelling(Baseline Performance)

- We used a Convolutional Neural Network (CNN) to identify patterns in leaf images.
- The model was trained using 70% training, 15% validation, and 15% testing data.
- It achieved an accuracy of 96% during testing.



Transfer Learning Model

- Tried MobileNetV2 for better accuracy.
- Transfer learning improved generalization but required fine-tuning.
- Best performance: around 92% accuracy after tuning.
- Model was saved and deployed for testing via Streamlit.



Recommendations



- Continue improving the model's accuracy using advanced AI architectures and a larger, balanced dataset.
- Collect more real-world images from farms.
- Improve app usability for mobile access.
- Partner with agricultural experts to validate results.
- Educate farmers on capturing quality images for testing.



Conclusion

- Automated plant disease detection is feasible and highly effective using CNNs.
- The model achieved high accuracy and can assist farmers with early detection.
- Reduces reliance on experts and enables faster response to disease outbreaks.
- Supports sustainable and data-driven agriculture.



Next steps

- Test model with real farm data.
- Retrain using more balanced datasets.
- Explore other models like ResNet or EfficientNet.
- Integrate recommendations and feedback into app updates.



Deployment

- Deployed using Streamlit for easy access.
- Users can upload a leaf image and receive instant predictions.

Below is the deployment link

Click here to deploy





Thank You!

Presented by Leaf it to us